

NETWORKED CONTROL

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EXERCIZES

1) Consider the following continuous-time system $\mathcal{S}$.

$$\dot{x}(t) = (\log_e 2)x(t) + \log_e 4 u(t)$$

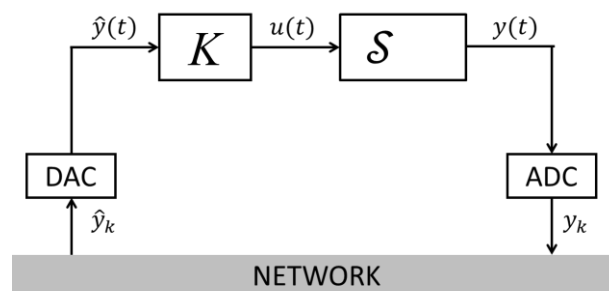
- A. Compute the eigenvalue and the mode of the system. Is the system open-loop asymptotically stable?
- B. We aim to control the system using a digital quantized controller. Answer the following questions.
 - a. According to the theory, what is the minimum rate N_B/h (where N_B is the number of bits encoding the control law and h is the sampling time) required to make the system boundable?
 - b. In view of the answer given to the previous question, what is the maximum value of h that can be used to make the system boundable in case $N_B = 2$?
 - c. Set $h = 1$ s and $N_B = 2$. Using the control law defined in case we operate at the rate limits (with $u_{\min} = -1$ and $u_{\max} = 1$), analyze the properties of the state trajectories using the graphical method (possibly starting by different selected initial conditions). Compute the corresponding invariant set.
 - d. Is the control law used at point c. the unique one that allows to steer the state to lie in a bounded set? **Justify adequately the response.**

2) What are the main **differences between MAC protocols** based on **scheduling** and the ones based on **random multiple access**? **Summarize their main features, advantages and disadvantages** and **list some main protocols** that belong to these two classes.

3) Assume that the linear system

$$\begin{cases} \dot{x}(t) &= Ax(t) + Bu(t) \\ y(t) &= Cx(t) \end{cases}$$

is controlled through the **networked control system (NCS)** sketched in the following figure.



Assume that the **sampling time** h_k and the **delay** τ_k affecting the control network are **time varying** and that $h_{\min} \leq h_k \leq h_{\max}$, $\tau_{\min} \leq \tau_k \leq \tau_{\max}$.

A. Define the **dynamic model describing the NCS dynamics**.

B. Answer the following questions.

- What is the **class** to which the model of the NCS belongs?
- What are the challenges related to the **stability analysis** of such system with respect to classical linear time invariant systems?
- State the **condition that guarantees the stability of the NCS**. Please justify the latter condition in the light of the **Lyapunov stability theory**. Is it **necessary and sufficient**? Why?